FREE PREVIEW

MX:
THE INTRO

Chapter zero of the guide to building a web that works for AI agents — and everyone else

CONTINUE WITH

MX: THE HANDBOOK

Everything you need to design the web for AI agents — and everyone else

MX: THE PROTOCOLS

The formal standards and specifications for an AI-readable web

This is not the full system.
It's the starting point.

Unlock the full system →

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More AI traffic

More conversions

More revenue

Future-proof your business

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MX: The Introduction

Understanding the machines reading your website, and the revenue at stake.

Revenue you cannot see leaving

Machines are mediating commerce right now. They compare your products, check your availability, and recommend you or your competitors. The commercial question is straightforward: are they completing transactions on your site, or moving on to someone who made it easier?

Adobe's Holiday 2025 data makes the scale visible. AI referrals to retail sites surged by 700%. Travel referrals rose by 500%. Conversion rates from AI-referred visitors now lead human traffic by 30%, and those visitors are 33% less likely to bounce. Machine-mediated commerce moved from experimental to revenue driver in a single quarter.

In January 2026, four major platforms launched machine commerce systems within eight days: Amazon Alexa+ (5 January), Microsoft Copilot Checkout (8 January), Google Universal Commerce Protocol (11 January), and Anthropic Claude Cowork (12 January). What industry analysts predicted would take 12–24 months compressed to six months. Machine-mediated commerce is infrastructure, not experiment.

The machines want to buy from you. The question is whether your content gives them what they need.

Your website is a fraction of your content estate. Contracts, policy documents, product specifications, and technical reports do not live on the web — but AI agents inside enterprise tools are already reading them, inferring what they can, guessing the rest. Being on the web and being machine-readable are not the same thing. A document on a public server is still opaque to an agent that has no context about what it is, who wrote it, what version it represents, or what claims it makes. Web accessibility solves discoverability. MX solves comprehension.

The invisible users

There is a new class of user on your website, and they are called invisible for two reasons. They are invisible to you: they blend into analytics, arrive once, and leave no trace. The interface is also invisible to them: they cannot see animations, colour, toast notifications, loading spinners, or any of the visual cues your site uses to communicate. These are machines visiting your website and performing actions without your awareness.

Most companies do not track AI bot traffic. Some block it entirely through robots.txt directives or Cloudflare identity checks, whilst simultaneously wondering why their AI-referred revenue is lower than competitors'. Modern AI browsers do identify themselves in their User-Agent strings, but those strings are trivially spoofed by any developer. You cannot know which type of machine is visiting, or how capable it is.

The patterns MX requires for these invisible users (explicit structure, semantic HTML, persistent state) also happen to benefit users with disabilities who rely on screen readers and keyboard navigation. That convergence is real. The business case for action is what drives the investment: machines are mediating commerce, answering health questions, comparing universities, and completing government transactions on behalf of humans. When your site fails them, you lose that business silently.

The convergence extends to the documents you publish, not just the pages. A typical organisation ships PDFs alongside its web content: white papers, product datasheets, regulatory filings, contracts, training materials. Most of those PDFs are byte streams without structure tags or alt text, which makes them unreadable to screen-reader users (one in four people in the European Union has some form of disability) and also poorly extractable by machines — AI agents, search-engine crawlers, and every automated system that reads documents. The European Accessibility Act, in force from 28 June 2025, applies financial penalties of approximately ten thousand euros for minor violations and one hundred thousand euros or more for major or repeated non-compliance, with significantly higher figures available against large enterprises. Producing tagged, declared, verifiable PDFs is the same work as producing accessible web pages: tag the structure, declare the conformance, record the check. MX makes the resulting documents accessible as a side effect of making them machine-readable, exactly as it does for HTML.

This is also where UNESCO's fairness principle lands. The Recommendation requires AI systems to respect fairness, non-discrimination, and digital literacy. MX's WCAG requirement and its insistence on existing primitives — Schema.org, semantic HTML, ISO formats, BCP-47 language tags — are the technical expression of that requirement. All users, regardless of disability, language, or device, deserve content that works for them. Accessibility is the visible edge of that commitment; machine-readability is the other edge.

Three failures the web cannot fix itself

Three interlocking structural failures block machine access, and no amount of AI improvement can resolve them.

Hostile design. Every website demands something before delivering value: accept cookies, dismiss the banner, close the pop-up, navigate the carousel. A toast notification that vanishes after three seconds is invisible to software reading the DOM on a half-second cycle. Tabs hide content the machine never requested. Accordions and show-more toggles collapse it before the machine arrives. The hostile web is architecturally incompatible with machine access, and machines will not get better at reading it. The ambiguity in your markup is permanent until you remove it.

Content imprisoned in platforms. Every CMS traps content in its own format, its own database, its own API. Machines cannot read any of it, because the content was designed for human eyes inside a proprietary system, not for machines operating across open standards.

Structured data is not enough. Schema.org describes. MX executes. Schema.org JSON-LD addresses what machines can *read*, but not what they can *do*. It does not fix hostile design. It does not free content from lock-in. It does not govern who may access the content, under what terms, for what purposes. Structured data is one layer in a practice that must span the entire interaction surface.

MX is not Generative Engine Optimisation. GEO — the vendor category built around optimising content so LLMs cite it — asks one question: how do I get an AI to mention my page? MX asks a different question: can any machine, in any format, at any time, find any document in a corpus, verify it is genuine, and know whether it is current? GEO optimises a single pathway. MX builds coverage across every pathway, because which pathway any given machine will take is unknowable in advance. Implement MX and GEO improvement follows automatically. The reverse is not true.

The diversity matters in practice. There is no single "LLM" to optimise for: ChatGPT, Claude, and Gemini have different ingestion paths, different fetch behaviour, and different trust signals. A tactic that lifts your citations in one can do nothing in another, and the gap widens every time a vendor tightens its rules, changes its system prompt, or ships a new model. New readers arrive every quarter — agents you have never heard of, parsing your pages by rules you cannot predict. The plumbing is what carries you across that churn: right MIME types, a discoverable sitemap, an llms.txt that is actually wired up, structured data that matches what is on the page. Four items, each a configuration change in tools you already use. Tactics will reshuffle every few months. Plumbing does not.

Machine Experience (MX) addresses all three — and it is not an internet problem. It is a document problem. Contracts, manufacturing specifications, government archives, medical protocols, scientific papers: the substrate of every enterprise. The same structural failures that block machines on a retail website block robots reading maintenance manuals, autonomous vehicles parsing safety procedures, and industrial systems consuming manufacturing specifications. MX covers the entire asset space. Every document any machine might encounter — in any format, on or off the internet — is within scope. The document is the problem. The document is also the fix.

MX sits alongside UX, SEO, and accessibility as a fourth discipline in web practice, and it shares the same foundational principle Steve Krug applied to human usability: don't make me think. When a machine has to think (inferring meaning, guessing formats, imagining context that was never written down) you have failed at MX. The difference is that a frustrated human visitor might complain, come back, or call your support line. A machine says nothing. It routes to a competitor whose HTML is easier to parse, and you never know it happened.

One principle underlies the practice: design for the most constrained machine and every machine benefits. You cannot know which system is consuming your document — user-agent strings are easily spoofed, and the range of machines reading documents now extends far beyond web browsers and AI assistants to robots, industrial systems, and devices that have no concept of a visual interface at all. Design for the worst and every system gains.

The common objection is that AI will get better and eventually handle poor markup. The precise version of that objection fails in a specific way: hallucination is an infrastructure problem, not a model problem. Better models hallucinate with more confidence on ambiguous data, not less. Foundation models will improve. The trend also runs the other way: small language models are coming to every phone, industrial controllers are consuming documents in real time, and the class of machines that need to read your content is expanding faster than any single model can accommodate. The most constrained machine is not a historical edge case; it is the near-term majority. Designing for it is not a hedge; it is preparation for what is already arriving.

Four commitments follow from that principle:

- **Typed over prose.** Prices, dates, identities in typed fields with agreed formats (ISO 4217, ISO 8601, BCP-47). Prose alongside, for humans.
- **Bound, not inferred.** A number on a page belongs to the entity it describes. Structural binding stops the machine guessing which entity a value applies to.
- **Human always in the loop.** Outputs remain readable and reviewable by both. Anything legible to only one reader (human or machine) is a regression. UNESCO's seventh principle — human oversight and determination — says the same thing in policy language: if no human can review what a machine did, the principle has already failed.
- **Build on what is already known.** Schema.org, JSON-LD, microdata, semantic HTML. Existing primitives, used with discipline.

When it goes wrong

A buyer decides to purchase a Kindle Scribe Colorsoft from Amazon at £570, credit card ready. The UK listing says "coming soon." His AI assistant can find the product and quote the price, but cannot complete the purchase, check stock across regions, or tell him when to come back. The machine could not finish the transaction. The revenue went nowhere. Amazon made it surprisingly difficult to take his money.

In January 2026, two high-speed trains collided near Adamuz in Spain. Forty-six people died. Within hours, pricing algorithms at airlines and car hire firms detected surge demand and raised prices by up to 200%, whilst families scrambled to get home. The algorithms worked correctly. Demand up, supply down, price up. Nobody had given them the governance rules that would have recognised a national tragedy. Spain's national ombudsman launched an investigation. The reputational damage will outlast the revenue gained.

Both failures share one root cause: machines acting on incomplete data. Commerce loses revenue. Public services lose trust, and sometimes lives. The technical solution serves both.

Why this matters to you

Whether you sell products, inform patients, publish research, or guide citizens through government services, machines need explicit structure to complete those actions.

A machine cannot reliably extract your price because "€2.030,00" could be two thousand and thirty euros or two point zero three. It cannot find your stock status because it lives in a JavaScript variable the DOM never exposes. It cannot complete your checkout because the "Confirm" button is a `<div>` with no semantic meaning. It skips you in recommendations and you never see it leave.

These are standard patterns on millions of websites. Every one of them breaks for automated visitors.

Consider what that looks like in practice. A buyer asks their shopping agent to find noise-cancelling headphones under £200. Two competing sites appear in the agent's crawl. The first has prices behind "See pricing" buttons, specifications locked in image-based charts, and stock status behind a login gate. The agent reports back: "I found several options but couldn't verify current prices or availability." The second has prices in Schema.org `Offer` markup, specifications in structured lists, real-time stock in the `availability` property. The agent reports: "Product X at £179, 4.7 stars from 2,800 reviews, ships in two days. Shall I proceed?" The second site gets the sale. The first was never considered.

The failure is silent. When an agent cannot read your page, it does not send a complaint. It routes to a competitor whose HTML is easier to parse. No analytics signal. No error report. Quiet abandonment at the moment of intent, and you will not find it in your dashboards.

When an agent cannot read your page accurately, it does not stay silent — it confidently tells the user what it thinks it found. "Your AI said the price was £49." "ChatGPT told me you were open on Sundays." "The agent said you ship to Canada." Every ambiguity in your HTML becomes a support ticket, multiplied across every user who trusted the answer. Structured data is a customer service investment as much as a discovery one.

The scale is measurable. Most sites lack proper semantic HTML. The majority have no llms.txt. More than half have partial or missing Schema.org structured data. A significant proportion block major AI crawlers outright, whilst simultaneously wondering why their AI-referred traffic lags behind competitors'. The infrastructure the machine economy expects simply is not there.

Check your own server logs today. Search your access logs for GPTBot, Claude-Web, PerplexityBot, and Bytespider. These visitors arrived, read your content, and either used it to answer someone's question about your business, or failed and moved on to a competitor. You have no visibility into which outcome occurred unless your content is structured for machines to read accurately. This diagnostic takes five minutes. The answer tells you whether MX is urgent or merely important.

How machines actually access your site

There are two fundamentally different mechanisms of machine access, and they operate on completely different timescales.

Training-time ingestion: historical knowledge building

During model training, large language models ingest web content through datasets like Common Crawl, massive archives of crawled websites that form the knowledge base machines draw upon when answering queries. This ingestion happens months or years before machines interact with users, making it fundamentally historical rather than real-time.

Common Crawl operates as a web archive service that periodically crawls public websites, respecting robots.txt directives and sitemap.xml files. The crawled content becomes part of training datasets that teach models about the structure, patterns, and information available on the web. When machines reference your product specifications or company information without visiting your live site, they're drawing from this historical knowledge base acquired during training.

This training-time access has specific characteristics. It's indirect: your website doesn't interact with the model; it interacts with crawlers that build datasets used later for training. It's historical: the content the model learns from could be months or years old by the time users interact with the trained model. Crucially, it respects robots.txt: ethical crawlers honour your access control directives.

Inference-time access: real-time direct interaction

During user queries, machines may directly access your website in real time. When someone asks ChatGPT "What laptops does Example Store currently offer?", the system might fetch your live website using browser automation, API calls, or direct HTTP requests. This is inference-time access, software retrieving current information whilst processing a specific user query.

This direct access operates differently from training-time ingestion. It's direct: the machine fetches your website whilst the user waits. It's real-time: the content retrieved is your current live site, not a historical snapshot. It's specific: the system targets particular pages relevant to the query, rather than crawling your entire site. Critically, it may not respect robots.txt: software executing user queries might access content regardless of crawl directives, treating robots.txt as guidance rather than absolute constraint.

There is a further distinction most businesses miss entirely. There are two states of any web page: the served HTML (the raw markup delivered from the server, before any JavaScript has run) and the rendered HTML (what appears in a browser after the DOM has been fully manipulated). Most businesses test only the rendered state. Server-side machines, including the crawlers that build training datasets and the agents that answer live queries, receive the served HTML. If your prices, stock status, and product specifications live in JavaScript variables, those machines see none of them.

The distinction matters for authentication and authorisation. Content blocked by robots.txt won't enter training datasets through Common Crawl, but inference-time machines actively executing user queries might access that content anyway. If your content truly must remain private, authentication mechanisms (login requirements, API keys, IP restrictions) provide reliable access control whilst robots.txt alone does not.

Why both mechanisms need the same fix

The good news: the same MX patterns serve both mechanisms simultaneously.

Sitemap.xml serves both. Semantic HTML serves both. Schema.org structured data serves both particularly well; during training, JSON-LD blocks teach models about entity relationships and product attributes; at inference time, machines parse that same JSON-LD to extract current pricing and availability without interpreting prose.

There is one notable exception. The llms.txt file has a structural problem that limits its reach. It is served as a text or markdown MIME type, not HTML. Common Crawl ingests HTML pages; it does not typically ingest non-HTML files. Training-time crawlers may never discover it. At inference time, automated systems typically do not fetch it either; a system answering "What laptops does Example Store sell?" goes straight to product pages, not a site-level directory file. To close this gap, publish the same content as an HTML page and include it in your sitemap.

The Gathering (the independent standards body that governs MX specifications) addresses this in the MX Agent Directory Discovery note. The note is a draft standard that applies to any agent-directory file your site might publish, llms.txt today, ai.txt or whatever follows tomorrow. It defines three things any organisation can verify in an afternoon. First, transport: serve the file as `text/html` so the largest training crawlers can ingest it. Second, discovery: list the file in your `sitemap.xml`. Third, resilience: include a `<link>` tag in every page that points to the file, so the reference survives even when your page body is built by JavaScript and is empty for the crawlers that do not run scripts. None of the three requires a new piece of infrastructure. Each is a configuration change in tools you already use. The note exists because none of the three is what most teams currently do, and the file is invisible until all three are in place.

Training vs learning vs codification

An important distinction: machines do not get "trained" when they visit your site. Their neural network weights are fixed. What actually happens involves three distinct layers:

Layer	What Happens	Cost	Persistence
Training (AI company's work)	Models learn general capabilities from datasets like Common Crawl: HTML syntax, Schema.org patterns, structured data extraction	\$50–100M per model (one-time)	Permanent but historical
In-context learning (per session)	Machine reads your HTML, understands your specific patterns, navigates your site	\$0.10–1.00 per session	**Disappears when session ends**
Codification (what MX creates)	Semantic HTML and Schema.org markup capture knowledge permanently in the page itself	Time to implement (one-time)	**Permanent and reusable**

Without MX, every machine must re-learn your site structure from scratch every session: re-read, re-discover navigation, extract information, then forget everything. With MX, the structure IS the captured knowledge. These systems spend zero time learning and 100% of their processing capacity extracting accurate information. Implement once, every visitor benefits forever, saving compute and energy costs as well.

That saving is not incidental. UNESCO's Recommendation on the Ethics of Artificial Intelligence names environment and ecosystem flourishing among its four core values. One of MX's founding principles carries the same instruction: choose what is better for the planet — reduce compute, reduce inference, reduce energy. A machine reading a structured cog instead of guessing at unstructured prose pays a smaller energy bill, and that bill multiplied across every agent in the chain is real.

The machine journey

MX readiness is not binary. Machines attempt five sequential stages when they interact with a site, and failing at any stage blocks everything that follows.

Stage	Name	What it requires
1	Discovery	The site can be found: sitemap, robots.txt, llms.txt
2	Citation	Content can be identified and attributed: type, author, date, subject
3	Compare	Like-for-like comparison across entities: typed prices, availability, specifications
4	Price	Transaction-ready pricing: currency, validity, terms
5	Confidence	Trust signals: provenance, governance, authorisation

Most sites pass Stage 1 and fail Stage 2. Everything from Stage 3 onwards depends on Stage 2 working. An AI assistant cannot compare your products if it cannot first cite what those products are.

What MX-ready looks like

An organisation that implements MX reaches a measurable destination. Agents can find your content, identify it accurately, and cite it without inference. Support queries that stem from agent misinformation decrease — fewer wrong answers means fewer tickets. Sales cycles that depend on machine-mediated comparison become faster, because agents can compare and commit without a human translating the page for them. Strategic assets (product knowledge, reviews, operational procedures) are sovereign and portable across any platform. You control what automated systems do with your content: you write the instructions, they follow them. No inference. No guessing. No dependence on AI getting smarter.

Consider what portability means in practice. If you have spent five years building reviews on a third-party platform and then migrate, every one of those endorsements stays behind. Zero reputation in the new system, despite years of earned trust. MX's Entity Asset Layer (covered fully in both books) changes this: your customer relationships, product knowledge, and brand signals become yours to carry, readable by any system, held by no single platform.

The window for first-mover advantage is measured in quarters, not years.

The trust layer: Reginald

Stage 5 of the machine journey — Confidence — sits apart from the others. Stages 1 through 4 are about machine-readability: can a machine find your content, identify it, compare it, and complete a transaction? Stage 5 asks a different question: can a machine *trust* what it found?

That question is becoming structural. As AI-generated content multiplies, a machine reading your contract, your product specification, or your policy document faces a provenance problem that better markup cannot solve. Who wrote this? When was it last updated? Is the version it is reading current, or superseded? Was it written by a human, an AI, or an automated system?

This is not a web problem. It is a document problem. Every organisation publishes documents alongside its web presence — technical references, compliance filings, standard operating procedures, research reports. Most carry no machine-readable provenance. An agent reading them guesses. Guessing is how a three-year-old policy gets cited as current. Guessing is how AI-generated content passes as editorial. Guessing is how trust erodes at scale, silently, without a complaint or a log entry.

Reginald addresses this. It is a public registry — the provenance layer for documents of any kind. When a document is registered, Reginald records a cryptographic signature: who published it, that it has not been modified since publication, and when the current version was issued. The claim is narrow and precise: *this is what the owner published, unaltered.* Reginald does not attest factual accuracy. It attests origin and integrity.

The scope extends to any document any machine might encounter. A markdown specification, a PDF, a policy file on a corporate server, a technical reference in a git repository. The cog format — a self-describing document with structured metadata — is the unit. Reginald is where cogs become discoverable and verifiable by any machine on earth, not just machines that already know where to look.

The practical effects compound. An agent that reads a provenance-attested document hallucinates less — it has verified origin and version to cite rather than gaps to fill by inference. Fewer inference steps means lower token consumption and lower energy draw. A chain of agents reading attested documents propagates verified provenance rather than re-deriving it at each step.

The regulatory dimension is arriving in parallel. The EU AI Act, the European Accessibility Act, and digital-records legislation across multiple jurisdictions are placing documentation, logging, and verifiability obligations on the organisations they cover: can you demonstrate that a document is what it claims to be, and that the content an AI acted on was genuine? MX and Reginald do not grant compliance with any of these regulations — that remains a legal duty of the organisation. What they do is make the documentation the organisation must produce structured, machine-readable, tamper-evident, and verifiable on request. Reginald's attestation, in particular — cryptographically verifiable, portable, and document-native — travels with the document and lets any reader confirm that document's provenance at any point in the chain.

MX and Reginald are the two pillars. MX makes content machine-readable. Reginald makes it machine-trustworthy. Stage 5 requires both.

That alignment has practical consequences. UNESCO's Recommendation arrives with two assessment instruments: an Ethical Impact Assessment (EIA) and a Readiness Assessment Methodology (RAM). Both require evidence — declared values, measured outcomes, auditable processes — and neither has anywhere to land until the systems they assess have a technical substrate to assess against. MX is one such substrate. A document with semantic structure, declared metadata, accessibility tagging, and Reginald-attested provenance gives an assessor concrete signals to score, rather than reasoning by inference. MX is not an ethics framework. It is the technical discipline that makes ethical assessment verifiable.

Where to go from here

I first understood this at CMS Kickoff 2024 in Florida. A speaker asked the room how an eight-year-old buys a toy plesiosaur. The child searches, points, wants. They do not navigate carousels, read brand copy, or create accounts. Machines behave identically, at billions of times the scale. That observation turned my understanding of AI upside down. AI is better suited for *consuming* content than creating it. The websites need improving, not the AI.

That insight became two books.

MX: The Handbook

Practical patterns, working code, and implementation guidance for developers, content teams, and the organisations that deploy them. Covers machine-readability (MX) and the document trust layer (Reginald) in full.

#	Chapter	What it covers
1	Don't Make AI Think	The worst-machine principle: why simpler HTML wins AI recommendations
2	How AI Reads	How agents fetch HTML, build a DOM tree, and extract meaning from structure
3	Guiding Principles	Four commitments: semantic clarity, structure, explicit metadata, redundancy
4	Content Architecture for Machines	Articles, products, FAQs, navigation, and why retrofitting always costs more
5	Metadata That Works	JSON-LD, Schema.org, Open Graph, and YAML frontmatter in practice
6	Navigation and Discovery	How machines find content before they can read it
7	The JavaScript Challenge	The rendering divide and its direct commercial consequences
8	Testing with Machines	Six MX readiness levels with a complete testing methodology
9	Common Anti-Patterns	Fourteen recurring mistakes, each with before-and-after code
10	Implementation Roadmap	A phased approach that delivers measurable ROI at each stage
11	Business Imperative	The chapter to share upward: evidence to justify the investment
12	Cogs and Reginald	Reginald as the provenance layer: every file as a cog, the public registry that attests origin and integrity

MX: The Protocols

The technical reference for architects and practitioners who need to understand the full scope of what machine experience means, and what to do about it. Covers both pillars: machine-readability (MX) and machine-trust (Reginald).

#	Chapter	What it covers
1	What You Will Learn	Orientation and scope
2	The Invisible Failure	Silent breakdowns: invisible form failures, vanishing errors, pixel-only state
3	The Architectural Conflict	Why modern web design is structurally hostile to machines, by design
4	Business Reality	The commercial case: what machine failure costs, and what readiness returns
5	The Content Creator's Dilemma	Machine-mediated discovery and advertising-funded publishing
6	The Security Maze	Machines inheriting authenticated sessions, acting on behalf of users
7	The Legal Landscape	Jurisdictions writing rules for a problem that has already scaled
8	The Human Cost	The digital divide and the machine divide: same causes, same fix
9	The Platform Race	Amazon, Microsoft, Google, Anthropic: machine commerce in seven days
10	Generative Engine Optimisation	Discovery patterns that work across SEO and GEO simultaneously
11	Designing for Both	Serving machines and humans from shared structural principles
12	Technical Advice	Working code, testing strategies, and implementation tools
13	What Agent Creators Must Build	The developer side of the machine relationship
14	Agent Protocols	Standards and patterns for agents visiting your site
15	Intent-Driven Publishing	Metadata-first architecture where structure drives content assembly

#	Chapter	What it covers
16	When Machines Remember	Sovereign, portable machine memory built on open standards
17	The Joymaker	A 1969 science fiction novel that predicted machine-readable architecture
18	Content That Manages Itself	Five generations of CMS, and why the current model is transitional
19	The Information Architecture of the LLM Web	A Three-Layer IA Framework synthesising everything
20	Cogs and Reginald	How Reginald establishes provenance and trust: the cog format, attestation, and the document registry for any machine on earth
21	The Fields and the Standards	Protocol proposals that give the architecture durability
22	Content Negotiation and the Markdown Trap	The format layer that determines what machines can access

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MX: The Handbook is available now. MX: The Protocols ships July 2026. Both share continuously updated appendices at https://mx.allabout.network/books/appendices/.

The standard itself is not mine to keep. The Gathering — the open, vendor-neutral standards body that governs MX — makes the rules and the enforcement of the rules visible to everyone who depends on them. That is UNESCO's fifth principle applied in practice: responsibility and accountability require governance any stakeholder can verify, not vendor decisions held in private. The Gathering's founding sponsors fund the work without owning it; the standard belongs to the community.

The structural failures described in this chapter do not fix themselves. They require deliberate work, and that work takes time. Starting before the wave breaks is not premature optimisation. It is the minimum responsible position.

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Continue reading with the full books.

MX: The Protocols — The complete technical reference: every pattern, every principle, every implementation detail for Machine Experience.

MX: The Handbook — The practical guide: audits, checklists, and step-by-step instructions to make your content work for every machine.

Purchase the books at: https://mx.allabout.network/books/index.html

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The AI Tipping Point: A Consultant's Takeaways from CMS Kickoff 2024

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As a first-time attendee at the Boye & Company event, I didn't know what to expect. But the speakers and content left me with a deeper understanding of the future of CMS – and how AI is changing everything.

Tom Cranstoun is the principal consultant at Digital Domain Technologies Limited and a CMS Critic contributor.

Reflecting on the Boye & Co CMS Kickoff 2024 down in St. Pete Beach, it's clear we're at a tipping point with AI in the CMS space. The event was a melting pot of CMS enthusiasts and experts, all there to predict and shape how we handle CMS. The setup was well-suited for learning, networking, and bouncing ideas off one another, with sessions covering a wide range of topics.

We had some standout talks, like Deane Barker's Q&A and Kristina Podnar's insights into Generative AI. Becky Brown and David Rasmussen hit home the importance of change management for content teams. Mike Wills' take on flexible content models for multichannel strategies was enlightening, and Jeff Eaton shed light on how CMS page builders affect team dynamics.

Brian Browning showed us practical AI applications that are changing digital experiences. Debbie Tucek pushed us to think of content as a product, while Ricky Frohnerath's masterclass on high-performance content teams was a game-changer. Nick Rudd's perspective on AI in real-time content generation was eye-opening, and sessions by Marli Mesibov and Brian McKeiver on the CMS buy vs. build dilemma and digital commerce were thought-provoking.

The twist in my expectations

I noted that the agenda of the CMS Kickoff 2024 offered only a *slight* focus on AI. The speakers often mentioned it during their talks; it's very "of the moment" in our industry.

I, like you, have been to many seminars, presentations, and conferences around AI and what it will do for you and your content. I was pleasantly surprised this was not about AI *creating* the content – it was about AI *reading* the content. It turned my expectations upside down, fired up my brain, and I could not stop taking notes.

The whole CMS Kickoff conference was not just about AI. I have many takeaways that I am working on. But the most important is this: *the AI revolution will change our industry and everyone else's*. Still, despite its great promise and clear benefits, it soon becomes clear that using AI to create content is not ideal. In fact, AI-generated content has many problems, including:

- Enforcing the use of internal style guides.
- Maintaining a brand tone of voice.
- Avoiding gender or race biases.
- Avoiding cultural nuances.
- Knowing which sources were used in the training of the AI (too few sources means limited vocabulary – conversely, too many sources suggest the corporation loses control. Does the AI manage to use local idiomatic expressions? For instance, a "sidewalk" is a "pavement" in the UK).
- AI sometimes gives up asking the prompter to continue.
- AI is becoming regulated, and legal ramifications will soon become commonplace, changing in each jurisdiction.

And this is just the beginning.

Rethinking AI's role in content management

Throughout the event, the speakers made me rethink AI's role in content management. Given the translation, bias, regulation challenges, and lack of trust in AI, one needs to employ reviewers and editors to correct the content – because AI is better suited for *consuming* content.

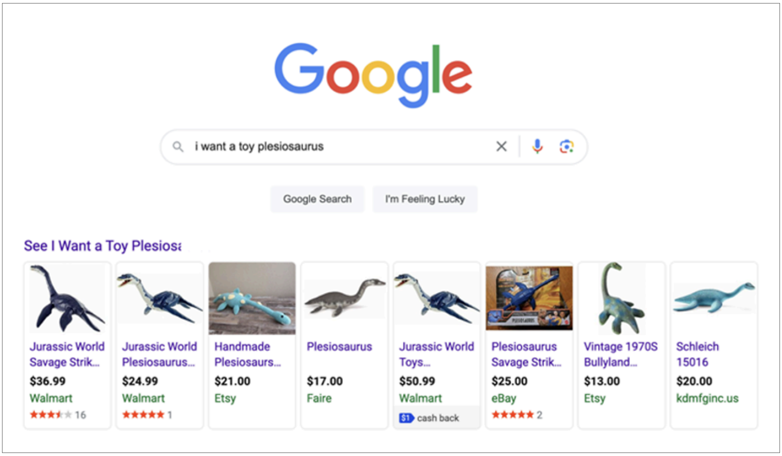
Let's think that the current state of AI creating content is a beta experiment, best left to the tech guys. After all, we have a business to think of.

Nick Rudd led an engaging discussion explaining how AI should be thought of. He began by asking how an eight-year-old purchases toys. He described it as simple and direct, without the baggage of brand loyalty or complex decision-making.

Kids are smart. They know what they want.

Ignoring the elephant in the room, "TikTok" (an entirely different article), the eight-year-old goes to Google/Bing/Whatever and types in:

"I WANT A TOY PLESIOSAURS"



Search results for "I WANT A TOY PLESIOSAURS"

What didn't the eight-year-old do?

- They didn't go to Amazon.
- They didn't refine the search.
- They didn't look at brands.
- They didn't read customer reviews.
- They didn't sort by price or popularity or whatever.
- They didn't read the price (well… mostly).
- They didn't create an account.
- They didn't click on dropdowns.
- They ignored video and animations.
- They didn't read the copy.

What happens when AI reads content? It behaves like an eight-year-old:

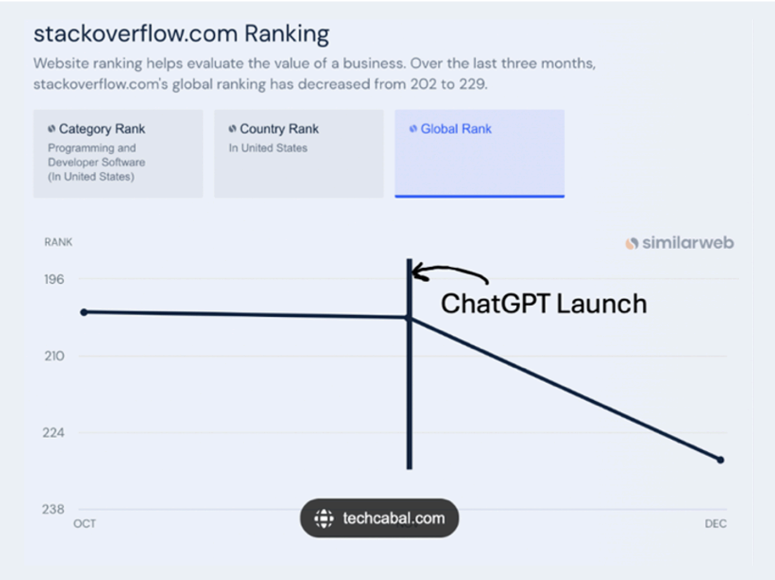
- It will ignore retailers.
- It will probably not refine the search.
- It will ignore brands.
- It might skim customer reviews.
- It might sort by price.
- It won't create an account.
- It won't follow your dropdowns.
- It won't give you an email address (you no longer know *who* your customers are).
- It will ignore the videos and animations.
- It will ignore the "see our shop links".
- It will skim-read the copy if it can find it among the ads.

How will “AI simplicity” affect your business?

The idea of simplicity will challenge how AI interacts with our current web designs and content structures.

The next step in AI monetisation is that the AI offers to *purchase* the toy – keeping any discounts, commissions, and recommendation fees bypassing the retailer. This will impact KPIs as AI activity increases and human interaction decreases.

What happened to StackOverflow recently will begin to affect us all. The online Q&A platform for developers is releasing 28% of its staff to save costs, which is not necessarily related to AI. But look at the point where ChatGPT entered the picture:



StackOverflow traffic graph showing decline after ChatGPT launch

What about a site’s KPIs?

As AI interaction increases, our reliance on traditional KPIs must fall.

Every website today uses analytics to drive statistics, such as:

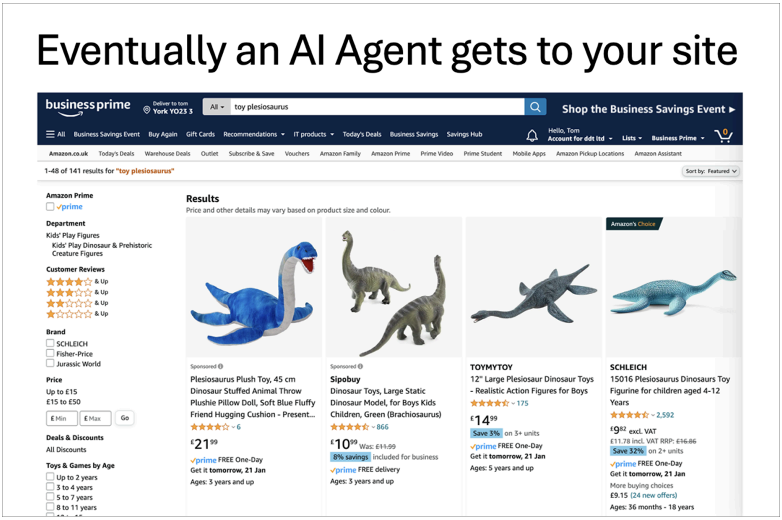
- The number of total visits.
- The number of unique visitors.
- The number of new vs returning visitors.
- Geographical location of visitors.
- Net promoter score.

- Pages viewed per session.
- Dwell time.
- Bounce rate.
- Conversions, including A/B testing.
- Organic search visitors.
- The number of social share impressions.
- Feedback and reviews.
- Cart abandonment.

Design by engineers: a trap

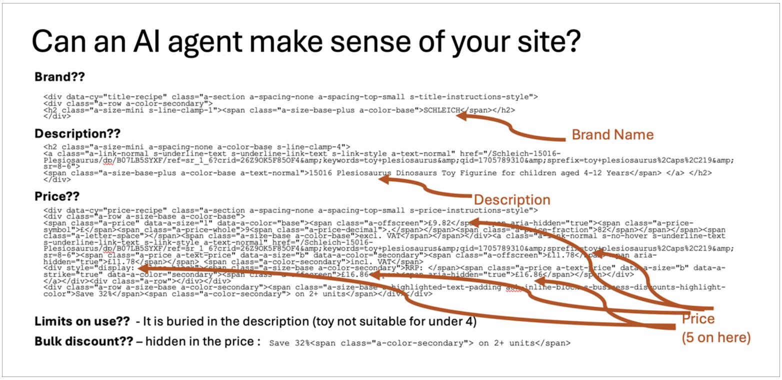
The conference also highlighted a common pitfall: websites designed by engineers focus on *looks* over *function*, which does not work well for AI. We need a more structured approach to content modelling for better clarity and organisation.

Today's HTML markup makes sense to the browser:



Clean product page rendered in browser

However, opening the "View Page Source" shows a different take:



Messy HTML source code of the same product page

Buried in the HTML are the brand name, the price (more than one is shown in the HTML), and the description. Note that discounts and age suitability are placed in the text. A human reading this would have difficulty, never mind an AI.

Engineers create text and image boxes in your CMS; they let you fill and serve them.

They might call them hero's carousels, cards, etc. They are dressed up text and image boxes with interactions.

This is the *Design by Engineer Trap*.

What needs to be added to our sites?

Content Modelling and schema presentation!

A lot has gone wrong. The IT people who made this website are very clever engineers; I could not do this.

The engineers took the site owners' desires and made pretty pictures with text and prices, highlighting discounts. They made it reasonably fast and effective and made the site owner one of the world's wealthiest men.

Look at other sites' HTML. How do they handle pricing? *Awful!*

Each site dresses up text in presentations. No semantic meaning is given to the objects described. There is no content modelling. It looks like AI will need an LLM for each brand!! This is not going to work in the AI world.

Something *major* must happen…

Content modelling the traditional CMS way

You probably already know it: the model gets built into the CMS.

This is a trap.

Designers fixate on the presentation when envisioning a website – and build a style guide with fonts, colours, components, and interactions. They create heroes, cards, lists, articles, carousels, accordions, tabs, text, and much, much more.

What next?

The content team is asked to create:

- **Testimonials.** *Someone decides that testimonials should be in the cards component.*
- **Special offers.** *Someone decides that testimonials should be in the lists component.*
- **A "coming soon" message.** *Someone decides that testimonials should be in the image with a link component.*
- **Bulk discount messages.** *Someone decides that Bulk Discount should be in the text component.*

Content modelling by developers!

So, what do we need to do about this?

We need technical improvements, enthusiastic use of content models, and application of schema without putting cognisant overload on the content team – which is a big ask.

I assure you it will be worth it.

Let's look at the HTML example we showed earlier, this time with schema applied:

```
<div itemscope itemtype="http://schema.org/Product">
<h1 itemprop="name">Plesiosaurus</h1>
<p itemprop="description">15016 Plesiosaurus Dinosaurs Toy Figurine.</p>
    <div itemprop="audience" itemscope itemtype="http://schema.org/PeopleAudience">
        <meta itemprop="suggestedMinAge" content="4">
        <meta itemprop="suggestedMaxAge" content="12">
        Suitable for ages 4-12.
    </div>
    <div itemprop="brand" itemscope itemtype="http://schema.org/Brand">
        <span itemprop="name">SCHLEICH</span>
    </div>
    <div itemprop="offers" itemscope itemtype="http://schema.org/Offer">
        <span itemprop="priceCurrency" content="GBP">Price: £</span>
        <span itemprop="price" content="11.98">11.98</span>
        <link itemprop="availability" href="http://schema.org/InStock" />In Stock
        <a itemprop="url" href="http://example.com/product">Product Page</a>
        <div itemprop="eligibleTransactionVolume" itemscope itemtype="http://schema.org/PriceSpecification"
        <meta itemprop="minValue" content="2">
        <span>Bulk discount for 2 or more units.</span>
    </div>
</div>
```

HTML source code with Schema.org markup applied

Now a machine can see the price, the description, the brand, the discount, and the age applicability. It knows it is for a product, not an ad on the site.

Call to action

We are in marketing, so we must end with a call to action. So here it is:

Our websites need to target four device types: *Mobile*, *Tablet*, *Desktop,* and *Machine* in the future. Our CMS strategy needs a solid content model that applies consistently across all platforms. This is key for targeted ads, personalised content, and campaign management.

However, the current developer-driven content modelling approach must be revised and rethought. We must wrap our content semantically and stick to structured data schemas. Schema.org is the place to start; that is the technical part. We as an industry need to evangelise and build a joint effort from designers, developers, and content creators to ensure content is built within the model and aligns with the established schema.

As such, teams need to add an "AI Evangelist" to the roles.

The AI Evangelist will be a key figure in our organisations, driving the adoption and implementation of AI technologies. This role bridges technical knowledge, strategic insight, and the ability to communicate AI's benefits to diverse audiences. The AI Evangelist will forge partnerships across design, development, content, and testing teams to craft a unified AI vision.

Key Responsibilities of an AI Evangelist:

- Champion AI technology adoption, showcasing tangible improvements to digital strategies and user experiences.
- Lead AI-centric models and strategies, focusing on predictive analytics, personalised content, and AI-powered search optimisation.
- Facilitate AI integration into workflows with cross-functional teams, ensuring smooth transitions and efficiency.
- Lead educational initiatives like workshops and training to build an AI-aware culture.
- Monitor and assess AI trends for strategic adoption.
- Cultivate a network with AI vendors and thought leaders to build expertise.
- Advise leadership on AI integration, securing necessary resources and investments.
- Examine new opportunities to decrease time-to-market and increase engagement.

Qualifications:

- Proficient in digital marketing, SEO, development, and UX design.
- Strong communicator, adept at explaining complex technology to varied audiences.
- Proven leader and collaborator in team-driven environments.
- A commitment to learning and staying current with AI developments.

Desired Skills:

- Background in digital analytics and personalised content strategies.
- Knowledge of CMS and digital content creation.
- Experience with app-based content platforms and audience engagement.
- Strategic thinker, ready to foster innovation and change.

Closing

CMS Kickoff 2024 was a significant event for me.

As an industry, we must reconsider our content strategies in the face of AI's evolution. As we gear up for AI's continued integration into our digital spaces, adopting a structured, semantic approach to content management is more important than ever in building cross-team strategies.

These reflections aren't set in stone but are a "call to arms" for the industry. Together, we must prepare for a significant shift in how we design, manage, and deliver content in an AI-first world.

Work with the Author

Tom Cranstoun has spent 25 years helping organisations get their content right: first for humans, then for search engines, and now for every machine that reads content. As Principal Consultant at CogNovaMX, he brings MX from theory to practice.

Consulting

A structured engagement to assess your current web estate and build an MX roadmap. We audit your pages the way agents see them: toast notifications, sliders, stepped workflows, hidden JavaScript variables, and every interaction pattern that breaks for machines, not just metadata and markup. We identify where revenue is leaking to competitors and deliver a prioritised implementation plan your team can execute immediately.

Training

A one-day hands-on workshop for your development, content, and product teams. Participants leave with practical skills: semantic HTML patterns, Schema.org implementation, agent testing techniques, fixing interaction patterns that fail for machines, and a working MX template for their own site. Available on-site or remote.

Speaking

Conference keynotes, lightning talks, and executive briefings on Machine Experience: what it is, why it matters commercially, and what organisations should do now. Tom has spoken at CMS Experts conferences in Frankfurt, Florida, and London.

Ready to make your web work for every machine on earth?

Email	info@cognovamx.com
Web	https://allabout.network
Books	*MX: The Handbook* · *MX: The Protocols*

Get in touch to discuss how MX can work for your organisation.

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The web is no longer consumed only by people.

AI agents are becoming a new consumption layer for enterprise digital platforms — interpreting, transforming, comparing, and acting on information on behalf of users, teams, and systems.

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Tom Cranstoun has shaped the technology industry for over 40 years, building products and systems used by millions. A long-standing member of the CMS Experts community, he has worked with organisations including Nissan, Ford, Jaguar Land Rover, and Twitter/X.

In 2024, his CMS Critic article identifying the "AI tipping point" reframed the conversation: designing for machines is now as important as designing for humans.

MX: The Handbook develops the full framework — the language, principles, and practical patterns for teams building the next generation of enterprise digital platforms.

Full release: MX: The Handbook
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This preview is the beginning.

MX: The Handbook provides the full framework — practical patterns, technical standards, and implementation guidance for anyone building or managing digital experiences in the agentic era.

If machines are reading your website, this book will change how you build it.

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